

EXERCISE 3: HYPOTHESIS TESTING

1. Weight change in Anorexic girls

A recent study compared different psychological therapies for teenage girls suffering from anorexic, an eating disorder that causes them to become dangerously underweight. Each girl 's weight was measured before and after a period of therapy. The variable of interest was the weight change, defined a weight at the end of the study minus weight at the beginning of the study. The weight change was positive if the girl gained weight and negative if she lost weight.

In the study, 29 girls received cognitive behavioral therapy, this form of psychotherapy stresses identifying the thinking that causes the undesirable behavior and replacing it with thoughts designed to help improve this behavior. The following weight data was collected:

Girl	Before	After	Girl	Before	After	Girl	Before	After
1	80.5	85.2	11	85.0	100.5	21	83.0	81.6
2	84.9	88.6	12	89.2	95.3	22	76.5	75.7
3	81.5	86.4	13	81.3	87.6	23	80.2	82.6
4	82.6	81.9	14	76.5	72.5	24	87.8	95.4
5	79.9	76.4	15	70.0	78.1	25	83.3	85.2
6	88.7	93.6	16	80.6	71.3	26	79.7	83.6
7	94.9	98.4	17	83.3	89.4	27	84.5	88.6
8	76.3	89.4	18	87.7	92.1	28	80.8	86.2
9	81.0	71.4	19	84.2	83.9	29	87.4	86.7
10	80.5	82.1	20	86.4	82.7			

Conduct a significance test for finding the strength of evidence supporting the effectiveness of the cognitive behavioral therapy, that is, to determine whether it results in a positive mean weight change for the significance level of 0.05.

Source: modified from Agresti, A., Franklin, C. and Klingenberg, B. (2018). Statistics: The Art and Science of Learning from Data, 4th Edition, Global Edition. Pearson.

2. Gum flavor longevity

In a test to determine the flavor longevity of a chewing gum, clients entering a store were asked to participate in an activity. The activity consisted of chewing a certain brand of gum and recording how long the gum flavor lasted in minutes. Records from groups of males and females were as follows:

Females:	15	21	29	22	19	25	35	23
Males:	22	24	23	30	12	17	28	

Use a statistical hypothesis testing to perform a two-sided significance test of the null hypothesis that the population mean is equal for two groups for the significance level of 0.05.

Source: Agresti, A., Franklin, C. and Klingenberg, B. (2018). *Statistics: The Art and Science of Learning from Data*, 4th Edition, Global Edition. Pearson.

3. Advertising claim

You and some friends have decided to test the validity of an advertisement by a local pizza restaurant, which says it delivers to the dormitories faster than a local branch of a national chain. Both the local pizza restaurant and national chain are located across the street from your college campus. You define the variable of interest as the delivery time, in minutes, from the time the pizza is ordered to when it is delivered. You collect the data by ordering 10 pizzas from the local pizza restaurant and 10 pizzas from the national chain at the different times. The data are as follows:

Local:	16.8	11.7	15.6	16.7	17.5	18.1	14.1	21.8	13.9	20.8
Chain:	22.0	15.2	18.7	15.6	20.8	19.5	17.0	19.5	16.5	24

At the 0.05 level of significance, is there evidence that the mean delivery time for the local pizza restaurant is less than the mean delivery time for the national pizza chain?

Source: Berenson, M., Levine, D.M., and Krehbiel, T.C. (2012). *Basic Business Statistics: Concepts and Applications*, 12th edition, Pearson.